

Chapter 2: Maps and Relations

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1. Maps

Let X and Y be two sets. A **map** f from X into Y is a process by which we associate a unique element of Y with every element of X . This means:

$$\forall x \in X, \exists! y \in Y, f(x) = y$$

- X is called the **domain** of f .
- Y is called the **codomain** of f .
- $f(x)$ is the **image** of x under f .

Let $f : X \rightarrow Y$ and $g : X' \rightarrow Y'$ be two maps. We say that f and g are **equal** if the following conditions hold:

$$X = X', \quad Y = Y', \quad \text{and} \quad f(x) = g(x) \text{ for all } x \in X.$$

In this case, we write $f = g$. For example, the following maps are not equal because their codomains differ:

$$\begin{array}{ll} f : \mathbb{R} \rightarrow \mathbb{R} & g : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0} \\ x \mapsto x^2 & x \mapsto x^2 \end{array}$$

Let $f : X \rightarrow Y$ be a map and let $A \subset X$. We define the **restriction** of f to A , denoted by $f|_A : A \rightarrow Y$, as the map defined by $x \mapsto f(x)$.

Let $f : X \rightarrow Y$ be a map and $B \subset Y$ such that for all $x \in X$, $f(x) \in B$. We define the **corestriction** of f to B , denoted $f'^B : X \rightarrow B$, as the map defined by $x \mapsto f(x)$.

1.1 Useful Maps

- **The constant map:** For any element $a \in Y$, the map is $f : X \rightarrow Y, x \mapsto a$.
- **The identity map:** $\text{Id}_X : X \rightarrow X, x \mapsto x$.
- **The projection map:** For a Cartesian product $X \times Y$:

$$\begin{array}{l} \text{Pr}_X : X \times Y \rightarrow X \\ (x, y) \mapsto x \end{array}$$

- **The indicator map.**

2. Images and Preimages of Sets

Let $f : X \rightarrow Y$ be a map and $A \subseteq X$. The set of all images of elements of A is called the **image of A under f** , denoted by $f(A)$. In other words:

$$f(A) = \{f(x) \mid x \in A\}$$

1. $f(\emptyset) = \emptyset$
2. $f(A) \subseteq Y$

Let $f : X \rightarrow Y$ be a map and $B \subseteq Y$. The set of elements in the domain whose images are in B is called the **preimage of B** , denoted by $f^{-1}(B)$.

$$f^{-1}(B) = \{x \in X \mid f(x) \in B\}$$

1. $f^{-1}(B) \subseteq X$
2. If $x \in f^{-1}(B)$, then $x \in X$ and $f(x) \in B$.

Consider the map $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $x \mapsto x^2$.

1. $f(\{1\}) = \{f(1)\} = \{1\}$
2. $f(\{-1, 1\}) = \{f(1), f(-1)\} = \{1\}$
3. $f([-2, 1]) = \{x^2 \mid x \in [-2, 1]\} = [0, 4]$
4. $f^{-1}(\{2\}) = \{x \in \mathbb{R} \mid x^2 = 2\} = \{\sqrt{2}, -\sqrt{2}\}$
5. $f^{-1}(\{-1\}) = \{x \in \mathbb{R} \mid x^2 = -1\} = \emptyset$
6. $f^{-1}([0, 1]) = \{x \in \mathbb{R} \mid 0 \leq x^2 \leq 1\} = [-1, 1]$

- Let $g = f|_{\mathbb{R}_{\geq 0}}$, i.e., $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$, $x \mapsto x^2$.

$$- g^{-1}([1, 2]) = \{x \in \mathbb{R}_{\geq 0} \mid 1 \leq x^2 \leq 2\} = [1, \sqrt{2}]$$

- Let $h : \{1, 2, 3\}^2 \rightarrow \mathbb{N}_0$, $(x, y) \mapsto x + y$.

$$- h(\{(1, 2), (1, 1), (2, 3)\}) = \{3, 2, 5\}$$

$$- h(\{1, 2, 3\}^2) = \{2, 3, 4, 5, 6\}$$

- $h^{-1}(\{7\}) = \emptyset$
- $h^{-1}(\{3\}) = \{(1, 2), (2, 1)\}$

3. Injection, Surjection, Bijection

Let $f : X \rightarrow Y$ be a map. We say that f is an **injection** (or injective or one-to-one) if distinct elements in X have distinct images in Y .

- For all $x_1, x_2 \in X$, if $x_1 \neq x_2$ then $f(x_1) \neq f(x_2)$.
- For all $x_1, x_2 \in X$, if $f(x_1) = f(x_2)$ then $x_1 = x_2$.
- For all $y \in Y$, the equation $y = f(x)$ has **at most one** solution $x \in X$.

A map f is not injective if and only if $\exists x_1, x_2 \in X$ such that $x_1 \neq x_2$ and $f(x_1) = f(x_2)$.

Let $f : X \rightarrow Y$ be a map. We say that f is a **surjection** (or surjective or onto) if every element of Y is an image of some element of X .

- For all $y \in Y$, $\exists x \in X$ such that $f(x) = y$.
- For all $y \in Y$, the equation $y = f(x)$ has **at least one** solution $x \in X$.

- A map $f : X \rightarrow Y$ is not a surjection if and only if $\exists y \in Y$ such that for all $x \in X$, $f(x) \neq y$.
- f is surjective if and only if $f(X) = Y$.
- f is surjective if and only if $f^{-1}(\{y\}) \neq \emptyset$ for all $y \in Y$.

Let $f : X \rightarrow Y$ be a map. We say that f is a **bijection** (or bijective) if f is both injective and surjective.

- For all $y \in Y$, the equation $y = f(x)$ has a **unique** solution $x \in X$.

4. The Inverse of a Map

Assume that $f : X \rightarrow Y$ is a bijective map. For every $y \in Y$, there exists a unique $x \in X$ such that $y = f(x)$. This allows us to define a map from Y to X , called the **inverse**

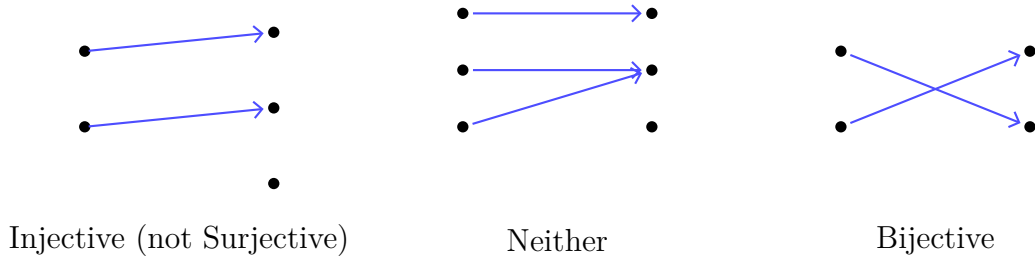


Figure 1: Visual representation of different types of maps.

map of f , denoted by f^{-1} .

$$f^{-1} : Y \rightarrow X$$

$$y \mapsto f^{-1}(y) = x$$

Then, for all $x \in X$ and $y \in Y$:

$$y = f(x) \Leftrightarrow f^{-1}(y) = x$$

Let $f : X \rightarrow Y$ be a bijective map. Then:

1. $f \circ f^{-1} = \text{Id}_Y$
2. $f^{-1} \circ f = \text{Id}_X$
3. f^{-1} is bijective.
4. If $g : Y \rightarrow Z$ is a bijective map, then $g \circ f$ is bijective, and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Let $L : (-\infty, 0] \rightarrow [0, +\infty)$ where $x \mapsto x^2$. The map L is bijective, so its inverse L^{-1} exists. The inverse is $L^{-1} : [0, +\infty) \rightarrow (-\infty, 0]$, where $y \mapsto -\sqrt{y}$. For any $x \in (-\infty, 0]$, we have

$$(L^{-1} \circ L)(x) = L^{-1}(L(x)) = L^{-1}(x^2) = -\sqrt{x^2} = -|x| = -(-x) = x$$

5. Binary Relations

Let X be a set. A **binary relation** R in X is any subset of X^2 . For $(x, y) \in R$, we write xRy . A relation R can have the following properties:

- **Reflexive:** For all $x \in X$, xRx .
- **Symmetric:** For all $x, y \in X$, if xRy then yRx .

- **Antisymmetric:** For all $x, y \in X$, if xRy and yRx , then $x = y$.
- **Transitive:** For all $x, y, z \in X$, if xRy and yRz , then xRz .

5.1 Equivalence Relations

An **equivalence relation** R on a set X is a binary relation that is **reflexive, symmetric, and transitive**.

Let R be the relation on $\mathbb{Z}$ where nRm if n and m have the same parity. This is an equivalence relation.

5.2 Equivalence Classes and Quotients

Let R be an equivalence relation on X and let $a \in X$. The **equivalence class** of a is the set $\bar{a} = [a] = \{x \in X \mid xRa\}$. The set of all equivalence classes is the **quotient set**, $X/R = \{\bar{x} \mid x \in X\}$.

Let R be an equivalence relation on a set X and let $a, b \in X$. The following are equivalent:

1. aRb
2. $\bar{a} = \bar{b}$
3. $\bar{a} \cap \bar{b} \neq \emptyset$

• Proof Sketch

- **(1 $\rightarrow$ 2):** Assume aRb . Use transitivity to show $\bar{a} \subseteq \bar{b}$ and symmetry plus transitivity to show $\bar{b} \subseteq \bar{a}$.
- **(2 $\rightarrow$ 3):** If $\bar{a} = \bar{b}$ and $\bar{a}$ is non-empty (since $a \in \bar{a}$), the intersection is non-empty.
- **(3 $\rightarrow$ 1):** If $x \in \bar{a} \cap \bar{b}$, then xRa and xRb . By symmetry, aRx . By transitivity, aRx and xRb implies aRb .

The quotient set X/R forms a partition of X .

6. Order Relations

A **partial order relation** on a set X is a binary relation that is **reflexive, anti-symmetric, and transitive**. It is often denoted by $\leq$.

6.1 Special Elements in a Poset

Let $(S, \leq)$ be a partially ordered set (poset) and $A \subseteq S$.

- An element $a \in S$ is a **minimal element** if there does not exist an element $b \in S - \{a\}$ such that $b \leq a$.
- An element $a \in S$ is a **maximal element** if there does not exist an element $b \in S - \{a\}$ such that $a \leq b$.
- An element $a \in S$ is the **least element** of S if $a \leq b$ for every element $b \in S$.
- An element $a \in S$ is the **greatest element** of S if $b \leq a$ for every element $b \in S$.
- An element $u \in S$ is an **upper bound** of A if $b \leq u$ for all $b \in A$.
- An element $l \in S$ is a **lower bound** of A if $l \leq b$ for all $b \in A$.

In a partially ordered set, the least and greatest elements of a subset are **unique**, if they exist.

Let $S = \{1, 2, \dots, 12\}$ with the relation $m \leq n$ if $m \mid n$.

- The elements 3 and 4 are **incomparable** since $3 \nmid 4$ and $4 \nmid 3$.
- 1 is the **least element** of S .
- The **maximal elements** of S are $\{7, 8, 9, 10, 11, 12\}$. There is no greatest element.
- Let $A = \{2, 5, 7, 10, 12\}$. 1 is a **lower bound** of A . A has no upper bound in S .

7. Countable and Uncountable Sets

A set S is **countable** if it is either finite or there exists a bijection $f : \mathbb{N}_0 \rightarrow S$. A set that is not countable is called **uncountable**.

1. $\mathbb{N}$ is countable via the bijection $f : \mathbb{N}_0 \rightarrow \mathbb{N}$ where $f(n) = n + 1$.
2. $\mathbb{Z}$ is countable. A standard bijection $f : \mathbb{N}_0 \rightarrow \mathbb{Z}$ is

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ -(n+1)/2 & \text{if } n \text{ is odd} \end{cases}$$

This maps $0 \mapsto 0$, $1 \mapsto -1$, $2 \mapsto 1$, $3 \mapsto -2$, and so on.